Equation of a tangent from a graph

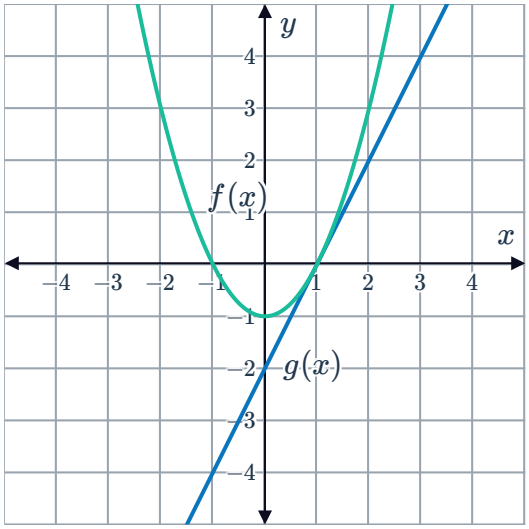
1 Each of the following graphs contain a curve, $f(x)$, along with one of its tangents, $g(x)$.

i State the coordinates of the point at which $g(x)$ is a tangent to the curve $f(x)$.

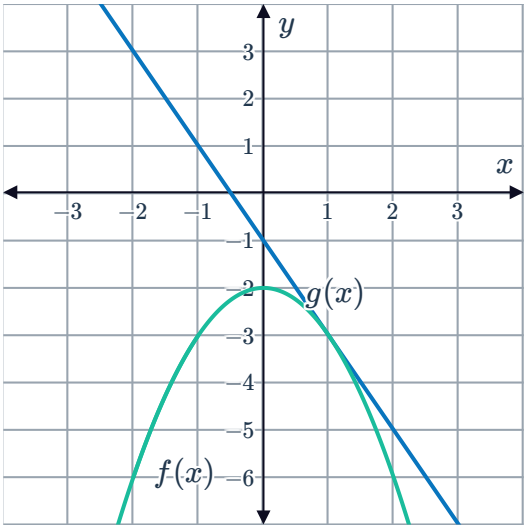
ii State the gradient of the tangent.

iii Hence determine the equation of the line $y = g(x)$.

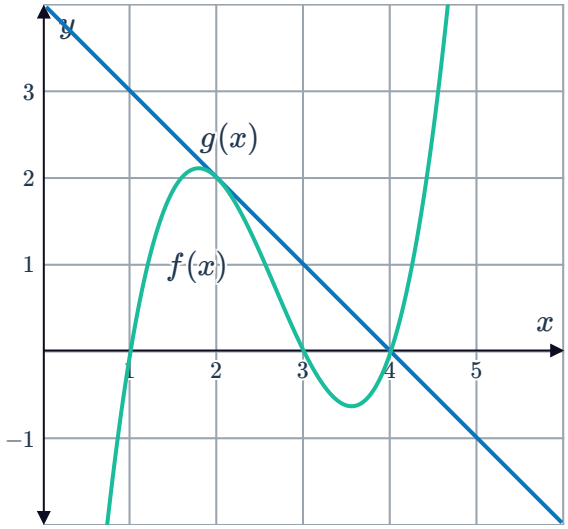
a



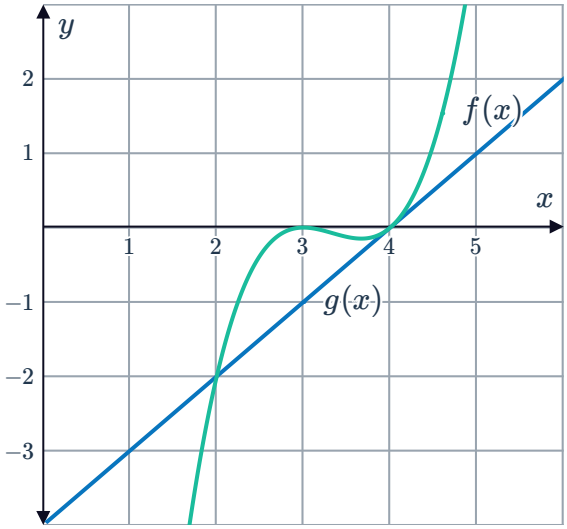
b



c

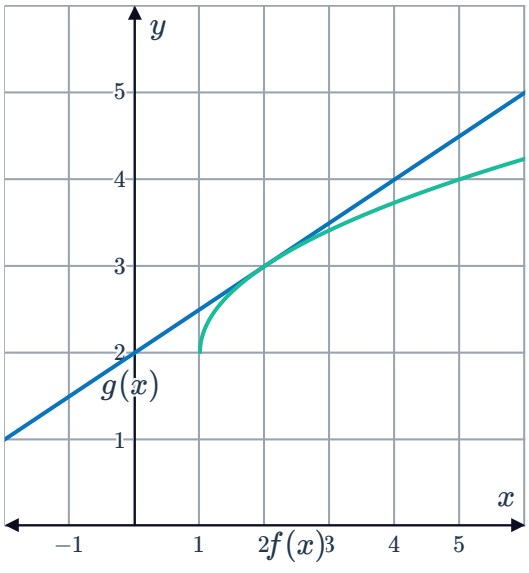


d



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e



2 Each of the following graphs contain a curve, $f(x)$, along with one of its tangents, $g(x)$.

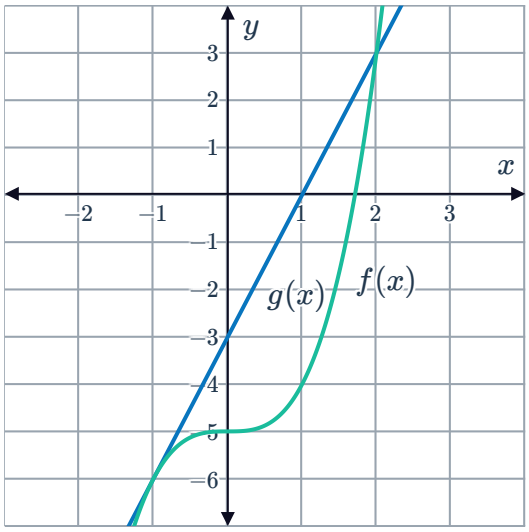
i State the coordinates of the point at which $g(x)$ is a tangent to the curve $f(x)$.

ii State the gradient of the tangent.

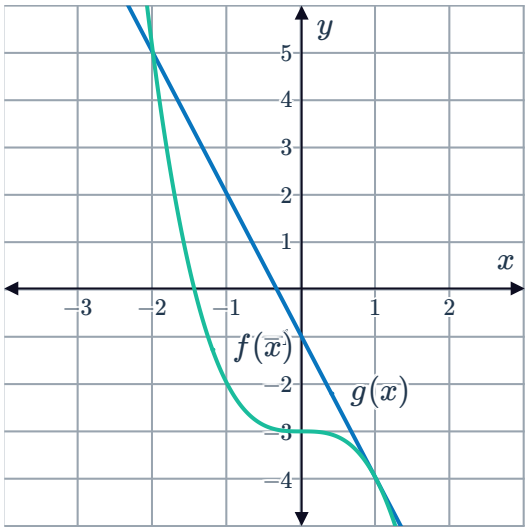
iii Hence determine the equation of the line $y = g(x)$.

iv State the x-coordinate of the point on the curve at which we could draw a tangent that has the same gradient as $g(x)$.

a



b

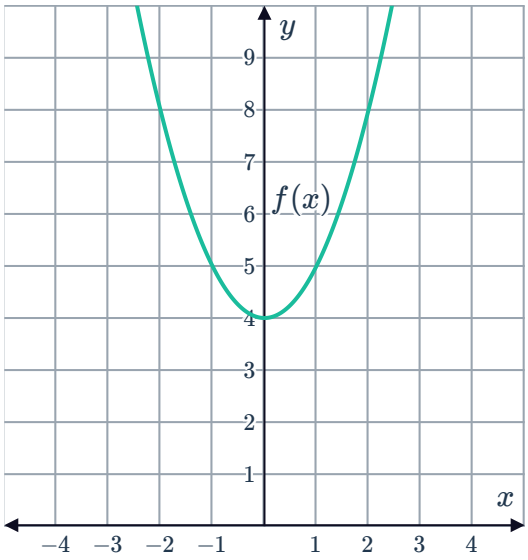


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3 Consider the graph of the function $f(x)$:

a Sketch the graph of the function $g(x) = 2x + 3$ on the same number plane.

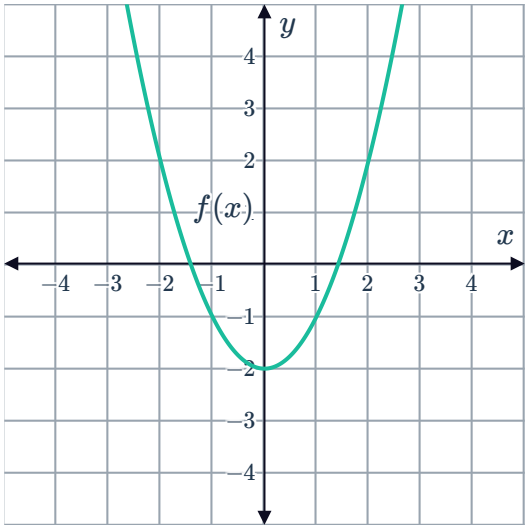
b Is $g(x)$ a tangent to $f(x)$? Explain your answer.



4 Consider the graph of the function $f(x)$:

a Sketch the graph of the function $g(x) = 2x - 1$ on the same number plane.

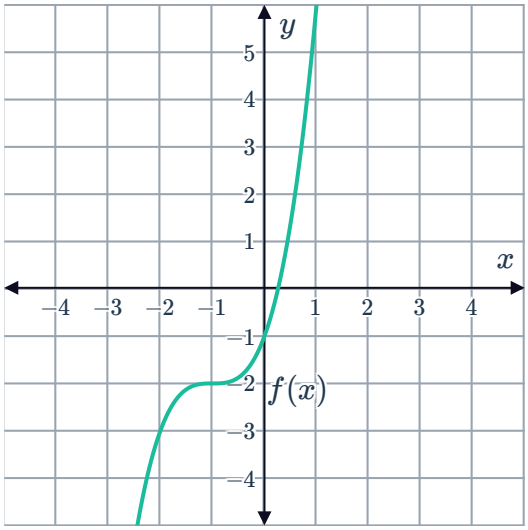
b Is $g(x)$ a tangent to $f(x)$? Explain your answer.



5 Consider the graph of the function $f(x)$:

a Sketch the graph of the function $g(x) = 3x + 3$ on the same number plane.

b Is $g(x)$ a tangent to $f(x)$? Explain your answer.

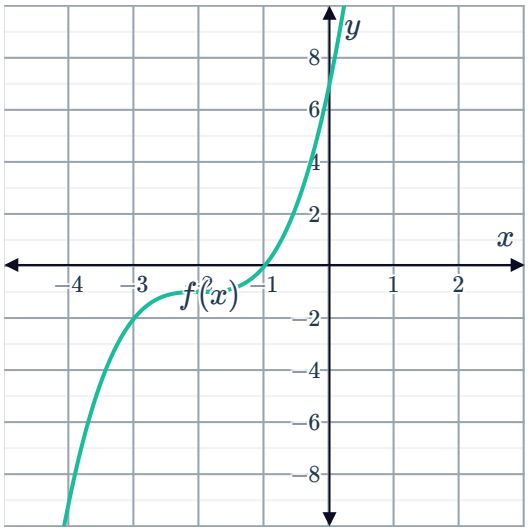


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6 Consider the graph of the function $f(x)$:

a Sketch the graph of the function $g(x) = 4x + 7$ on the same number plane.

b Is $g(x)$ a tangent to $f(x)$? Explain your answer.



7 Consider the curve given by the function $f(x) = x^2 - 1$.

a Find the gradient of the tangent to the curve at the point $(1, 0)$.

b Graph the curve and the tangent at the point $(1, 0)$ on a number plane.

8 Consider the curve given by the function $f(x) = x^2 - 4x + 2$.

a Find the gradient of the tangent to the curve at the point $(3, -1)$.

b State the coordinates of the vertex of the parabola $f(x) = x^2 - 4x + 2$.

c Graph the curve and the tangent at the point $(3, -1)$ on a number plane.

Equation of a tangent at a given point

9 Consider the function $y = x^2 - 3x + 4$. State the x-coordinate of the point on the curve where the tangent makes an angle of 45° with the x-axis.

10 Consider the tangent to the curve $f(x) = x^2$ at the point $(-1, 1)$.

a Find the gradient of the function $f(x) = x^2$ at this point.

b Find the equation of the tangent to the curve $f(x) = x^2$ at the point $(-1, 1)$.

11 Consider the tangent to the curve $f(x) = x^3$ at the point $(-1, -1)$.

a Find the gradient of the function $f(x) = x^3$ at this point.

b Hence find the equation of the tangent to the curve $f(x) = x^3$ at the point $(-1, -1)$.

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12 Consider the tangent to the curve $f(x) = -x^2$ at the point $(-3, -9)$.

a Find the gradient of the function $f(x) = -x^2$ at this point.

b Hence find the equation of the tangent to the curve $f(x) = -x^2$ at the point $(-3, -9)$.

13 Consider the tangent to the curve $f(x) = x^2$ at the point $(-2, 4)$.

a Find the gradient of the function $f(x) = x^2$ at this point.

b Find the equation of the tangent to the curve $f(x) = x^2$ at the point $(-2, 4)$

14 Consider the function $f(x) = 3x^2$.

a Find the gradient of the function at $x = 2$

b Find the y-coordinate of the point on the function at $x = 2$.

c Hence find the equation of the tangent to the curve $f(x) = 3x^2$ at $x = 2$.

15 Find the equation of the tangent to the curve $f(x) = 0.3x^3 - 5x^2 - x + 4$ at $x = 1$.

Applications

16 For each of the following:

i Find the x-coordinate of point M.

ii Find the y-coordinate of point M.

a At point $M(x, y)$, the equation of the tangent to the curve $y = x^2$ is given by $y = 4x - 4$.

b At point $M(x, y)$, the equation of the tangent to the curve $y = x^3$ is given by $y = 12x - 16$.

17 Consider the function $f(x) = \frac{4x^3}{3} + \frac{5x^2}{2} - 3x + 7$. Find the x-coordinates of the points on the curve whose tangent is parallel to the line $y = 3x + 7$.

18 $5x + y + 2 = 0$ is the tangent to the curve $y = x^2 + bx + c$ at the point $(9, -47)$.

a Find the derivative $\frac{dy}{dx}$ of $y = x^2 + bx + c$.

b Find the gradient of the tangent to the curve at $x = 9$.

c Solve for the value of b.

d Solve for the value of c.

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19 Consider the function $f(x) = x^2 + 5x$.

a Find the x-coordinate of the point at which $f(x)$ has a gradient of 13.

b Hence state the coordinates of the point on the curve where the gradient is 13.

20 Consider the function $f(x) = x^3 - 6x^2$.

a Find the x-coordinates of the points at which $f(x)$ has a gradient of 495.

b Hence state the coordinates of the points on the curve where the gradient is 495.

21 Consider the function $f(x) = x^3 + 3x^2 - 19x + 2$.

a Find the x-coordinates of the points at which $f(x)$ has a gradient of 5.

b Hence state the coordinates of the points on the curve where the gradient is 5.

22 Consider the function $f(x) = x^3 + 6x^2 - 14x - 2$.

a Find the x-coordinates of the points on the curve where the gradient is the same as that of $g(x) = 22x - 4$.

b Hence state the coordinates of the points on the curve where the gradient is the same as that of $g(x) = 22x - 4$.

23 Consider the function $y = 4x^2 - 5x + 2$.

a Find $\dfrac{dy}{dx}$.

b Hence find the value of x at which the tangent to the parabola is parallel to the x-axis.

24 The curve $y = ax^3 + bx^2 + 2x - 17$ has a gradient of 58 at the point $(2, 31)$. Find the values of a and b.

25 The graph of $y = ax^3 + bx^2 + cx + d$ intersects the x-axis at $(2, 0)$, where it has a gradient of 36. It also intersects the y-axis at $y = -28$, where the tangent is parallel to the x-axis.

Find the values of a, b, c and d.

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